

CASE STUDY ANALYSIS: ETHICAL CHALLENGES IN AI – POWERED LOAN APPROVAL SYSTEM



CHITKARA
UNIVERSITY

Submitted By –

Akhil Chopra (2520982511)

Niharika Gupta (2520982662)

Harshit (2520982584)

Submitted To –

Lavanaya Srivastava

Case Study: Ethical Challenges in AI-Powered Loan Approval System

1. Key Ethical Issues in the Case

The case highlights several important ethical concerns in the use of AI for decision-making:

- **Bias and Discrimination:** Female applicants and people from Tier-2 and Tier-3 cities were treated unfairly despite having similar financial profiles. This shows gender and geographic bias.
- **Lack of Transparency:** Customers were not given proper reasons for rejection, only vague responses like “system decision.”
- **Accountability Issues:** It is unclear who is responsible for incorrect or biased decisions—the bank, developers, or the AI system.
- **Unfair Decision-Making:** The system approved some risky applicants while rejecting deserving ones, indicating inconsistency.
- **Privacy and Data Misuse Risk:** Sensitive personal data was used without clear explanation of how decisions were made.

These issues violate principles of fairness, transparency, and ethical responsibility in AI systems.

2. Why Did the AI System Produce Unfair Outcomes?

The unfair outcomes occurred mainly due to **biased training data and poor system design**:

- **Historical Bias:** The AI was trained on past data that already contained human biases (e.g., preference for metro applicants or males).
- **Correlation vs Causation:** The model incorrectly assumed that gender or city is directly related to loan repayment ability.

- **Lack of Fairness Checks:** No mechanisms were implemented to detect or correct bias during or after training.
- **No Monitoring System:** The system was deployed without continuous evaluation of fairness across different groups.

As a result, the AI simply automated existing biases instead of eliminating them.

3. Is Model Accuracy Alone Sufficient?

No, **accuracy alone is not sufficient** to evaluate an AI system, especially in sensitive applications like loan approvals.

- A model can be highly accurate overall but still unfair to specific groups.
- For example, if most applicants are from metro cities, the model may perform well overall while discriminating against rural applicants.
- Ethical AI requires balancing **accuracy, fairness, transparency, and accountability**.

Therefore, success should be measured using multiple factors, not just accuracy.

4. How Can Explainability Improve Customer Trust?

Explainability means providing clear reasons behind AI decisions.

It improves trust in the following ways:

- **Transparency:** Customers understand why their loan was approved or rejected.
- **Fairness Perception:** People feel the system is fair when decisions are justified.
- **Opportunity for Correction:** Customers can improve their profiles if they know the reasons (e.g., low credit score).
- **Reduced Complaints:** Clear communication reduces confusion and dissatisfaction.

For example, instead of “system decision,” the system could say:
“Loan rejected due to low credit score and irregular income history.”

This makes the process more user-friendly and trustworthy.

5. Who Should Be Accountable for Wrong AI Decisions?

Accountability should be shared across multiple stakeholders:

- **Bank Management:** Responsible for deploying and approving the system.
- **AI Developers/Data Scientists:** Responsible for designing and testing the model.
- **Compliance and Risk Teams:** Responsible for ensuring ethical and legal standards.
- **Regulatory Authorities:** Responsible for enforcing fairness and transparency rules.

AI should not be treated as an independent decision-maker. Humans must remain accountable for its outcomes.

6. Suitable Fairness Metrics for This Case

To evaluate fairness, the bank can use the following metrics:

- **Demographic Parity:** Approval rates should be similar across gender and regions.
- **Equal Opportunity:** Applicants with similar repayment ability should have equal chances of approval.
- **Disparate Impact Ratio:** Measures whether one group is unfairly disadvantaged compared to another.
- **False Positive/Negative Rates:** Ensure errors are not higher for specific groups.

Using these metrics helps identify and reduce bias in the system.

7. Practical Steps to Improve the System

The bank can take the following steps:

a. Improve Data Quality

- Remove biased patterns from historical data.
- Include diverse and representative data samples.

b. Implement Fairness Techniques

- Use bias detection and mitigation algorithms.
- Regularly test the model for discrimination.

c. Add Explainability Tools

- Use tools like SHAP or LIME to explain decisions.
- Provide customers with clear reasons for outcomes.

d. Human-in-the-Loop System

- Keep human review for borderline or high-value cases.
- Avoid complete automation in sensitive decisions.

e. Regular Ethical Audits

- Conduct periodic audits to check fairness and compliance.
- Monitor performance across different demographic groups.

f. Transparency Policies

- Clearly communicate how AI is used in decision-making.
- Provide appeal mechanisms for rejected applicants.

8. Should Loan Approvals Be Fully Automated?

No, loan approvals should not be fully automated.

While AI can improve speed and efficiency, complete automation can lead to:

- Biased decisions
- Lack of accountability

- Reduced customer trust

A better approach is a **hybrid system**, where:

- AI handles initial screening and data analysis.
- Humans make final decisions, especially in complex cases.

This ensures both efficiency and fairness.

Conclusion

The Future Trust Bank case demonstrates that while AI can enhance efficiency, it also introduces serious ethical challenges if not properly managed. Bias in training data, lack of transparency, and absence of accountability led to unfair outcomes and loss of trust. To build a responsible AI system, the bank must focus not only on accuracy but also on fairness, explainability, and continuous monitoring. A balanced approach combining AI and human judgment is essential for ethical and effective decision-making.